

Solution

SOUND-02

Class 09 - Science

1. If the force has a component in the direction opposite to the displacement, the force does negative work.
2. Weight of the boy, $mg = 50 \text{ kg} \times 10 \text{ m s}^{-2} = 500 \text{ N}$
Height of the staircase, $h = 45 \times \frac{15}{100} \text{ m} = 6.75 \text{ m}$
Time taken to climb, $t = 9 \text{ s}$
power, $P = \text{Work done/time taken}$
$$= \frac{mgh}{t}$$
$$= \frac{500\text{N} \times 6.75\text{m}}{9\text{s}}$$
$$= 375 \text{ W.}$$

Power is 375 W.
3. When we wind a watch, the configuration of its spring is changed. The energy stored in the spring is obviously potential in nature (elastic potential energy to be more accurate).
4. When a force of 1N causes a displacement of 1m, in its own direction the work done is said to be one joule.
5. Two forms of mechanical energies are Kinetic energy and potential energy.
6. i) joule (ii) erg (iii) kilowatt hour
7. The person does no work because, no displacement takes place in the direction of applied force as the force acts in the vertically upward direction.
8. Work is done against the force of gravity.
9. $W = Fs = (5\text{N}) \times (2\text{m}) = 10\text{J}$
10. Yes we agree with her statement. Because when many balanced forces act on the object its displacement becomes zero.
11. Yes, when a force acts at an angle of 90° with the displacement.
12. It is the transformation of chemical energy of food to heat energy to our muscular energy to kinetic energy on paddling the bicycle.
13. Yes. For a uniformly moving object. If object is moving with constant velocity then inspite of zero net force acting on object there is displacement along the motion.
14. Power is defined as the rate of doing work. It is measured in watt and also in horse power.
15. A free falling object has maximum kinetic energy before hitting. Eventually on hitting ground it rolls and stops after some distance, since on striking the ground its kinetic energy is used to work against friction and kinetic energy is converted to heat energy.
16. Work done is given by the expression $W = FS \cos \theta$. When a body moves in a circular path, then the displacement (S) is Zero. Therefore, the work done is $W = F \times 0 = 0$
17. When a body does not perform any work, it never implies that the body has no energy. The body may have energy but still does not perform any work, e.g., a book placed on a table has potential energy but is not performing any work.
18. Mass of the object, $m = 15 \text{ kg}$, velocity of the object, $v = 4 \text{ ms}^{-1}$
$$K.E. = \frac{1}{2}mv^2$$
$$= \frac{1}{2} \times 15\text{kg} \times 4\text{ms}^{-1} \times 4\text{ms}^{-1}$$
$$= 120 \text{ J}$$

The kinetic energy of the object is 120 J
19. i) Chemical energy (ii) Heat energy (iii) Light energy (iv) Electrical energy (v) Sound energy (vi) Solar energy.
20. There is no work done as displacement is zero. While we push a huge rock with all our might(power) but fail to move it. energy transfer occurs and applied energy goes to rock and earth.
21. The capacity of a body to do work is called energy possessed by the body. It is a scalar quantity and is measured in joule (J).
22. Changing its velocity.
23. The Kinetic energy of the body will become one-fourth.
24. The energy possessed by a body by virtue of its motion is called kinetic energy.
25. The speed is doubled.
26. Work is said to be done when a force causes displacement of an object in the direction of applied force.

27. 1J is that work which is done when a force of 1N is applied on an object and object moves a distance of 1m in the direction of force.
28. Mass of the object, $m = 10 \text{ kg}$, displacement (height), $h = 6 \text{ m}$, and acceleration due to gravity, $g = 9.8 \text{ ms}^{-2}$.
 From given Equation,
 Potential energy = mgh
 $= 10 \text{ kg} \times 9.8 \text{ ms}^{-2} \times 6 \text{ m} = 588 \text{ J}$
 The potential energy is 588 J.
29. Zero
30. Work is said to be done if force acting on an object displaces it through a certain distance. It is measured as the product of force and displacement. Work done is negative if force and displacement are in the opposite direction.
31. Power of electric bulb = $60 \text{ W} = 0.06 \text{ kW}$.
 Time used, $t = 6 \text{ h}$
 Energy = power $\times$ time taken
 $= 0.06 \text{ kW} \times 6 \text{ h}$
 $= 0.36 \text{ kWh}$
 $= 0.36 \text{ 'units'}$.
 The energy consumed by the bulb is 0.36 'units'.
32. The winning team does work. The work is equal to the product of the resultant force and the displacement undergone by the losing team.
33. Mass of the object, $m = 12 \text{ kg}$,
 potential energy, $E_p = 480 \text{ J}$.
 $E_p = mgh$
 $480 \text{ J} = 12 \text{ kg} \times 10 \text{ ms}^{-2} \times h$
 $h = \frac{480 \text{ J}}{120 \text{ kgms}^{-2}} = 4 \text{ m}$
 The object is at the height of 4 m.
34. When a work of 1 joule is done in 1 s, the power is said to be one watt.
35. Mass of luggage, $m = 15 \text{ kg}$ and displacement, $s = 1.5 \text{ m}$
 Work done, $W = F \times s = mg \times s$
 $= 15 \text{ kg} \times 10 \text{ ms}^{-2} \times 1.5 \text{ m}$
 $= 225 \text{ kg ms}^{-2} \text{ m}$
 $= 225 \text{ N m} = 225 \text{ J}$
 Work done is 225 J.
36. $\text{Workdone} = \text{Force} \times \text{displacement}$
 $= 140 \text{ N} \times 15 \text{ m} = 2100 \text{ J}$
37. $\text{Power} = \frac{\text{Energy}}{\text{Time}} = \frac{1000}{10} = 100 \text{ w}$
38. i. a. As potential energy decreases with decreasing height, the speed of the object will increase and hence its kinetic energy will increase.
 b. Total mechanical energy will remain constant.
 It is based on the law of conservation of energy which states that energy can neither be created nor be destroyed/. It can only be transformed from one form to another.
- ii. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$
39. i. a. Kinetic energy of belt or the package.
 b. Heat energy
 c. Sound energy
- ii. $m = 36 \text{ kg}$, $h = 2.4 \text{ m}$, $g = 10 \text{ m/s}^2$
 $\text{G.P.E.} = m \times g \times h$
 $= 36 \times 10 \times 2.4$
 $= 864 \text{ J}$
- iii. $\text{power} = \frac{W}{t}$
 $\text{power} = \frac{864}{4.4}$

$$= 196.36 \text{ W}$$

iv. Mass is increased and power is constant, so increase in potential energy of mass is greater. Also, as mass is increased, speed is reduced and hence time taken by the conveyor is longer.

40. In Fig. (i), the angle between F and S is 90° , so work done is zero.

In Fig. (ii), the angle between F and S is 0° , so work done is positive.

In Fig. (iii), the angle between F and S is 180° , so work done is negative.

41. Weight of water displaced = 50 gf

Weight of alcohol displaced = (50 - 10) gf = 40 gf

As volume of water displaced = Volume of alcohol displaced

$$\text{RD of alcohol} = \frac{\text{weight of alcohol displaced}}{\text{weight of equal volume of water displaced}} = \frac{40}{50}$$

42. a. Promotes saving habits and judicious uses of resources and save resources

b. Joule.

- c.
- Switch off the power where it is not need.
 - Use of CFL light.
 - To check the proper wiring as to avoid leakage of power.

43. a. Potential energy.

b. If the stone is thrown without stretching the string of catapult the stone will fall down. As the stretched catapult possess potential energy due to stretch. It throws the stone with the high speed as string is released.

c. Shyam prevents his friend from aiming at the bird because stone could harm the bird and he does not want to harm or kill the bird, this shows his care and love for the living beings.

44. a. Sanchi explained that the energy required in both the cases is equal because whenever the work is done against gravity, the amount of work is equal to the product of weight of the body and the vertical distance through which the body is lifted.

(Work done in lifting a body = Weight of body $\times$ Vertical height).

b. Sanchi has problem solving values and conceptually strong.

45. i. Energy possessed due to the position of a body is called potential energy.

ii. The work done against gravity in both cases is mgh. It is independent of the path along which the body is moved and it depends only on the initial and final positions of the body.

46. Yes, Sony is right because if several forces acting on an object are balanced forces, i.e., their resultant is zero (i.e., $\Sigma F = 0$), then there is no acceleration of the object.

This may be seen as follows:

From Newton's second law, $\Sigma F = ma$

If $\Sigma F = 0$, $ma = 0$ or acceleration, $a = 0$

47. a. The passenger was right in saying so according to the laws of physics. Since the direction of force is vertically up and displacement horizontal, no work is said to be done.

b. Sympathy and value for other's work.

c. When you push a heavy boulder but are unable to displace it. (Any other situation)

48. $m_A = m_B = 1000 \text{ kg}$. $v = 36 \text{ km/h} = 10 \text{ m/s}$

Frictional force = 100 N

Since, the car A moves with a uniform speed, it means that the engine of car applies a force equal to the frictional force.

$$\frac{\text{Force} \times \text{distance}}{\text{time}} \text{ Power} =$$

$$= F.v$$

$$= 100 \text{ N} \times 10 \text{ m/s} = 1000 \text{ W}$$

after collision,

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$1000 \times 10 + 1000 \times 0 = 1000 \times 0 + 1000 \times v_B$$

$$v_B = 10 \text{ ms}^{-1}.$$

49. Displacement = 1500 m + 200 m + 2000 m = 3700 m

$$\text{Work done} = \text{Force} \times \text{displacement} = 5 \text{ N} \times 3700 \text{ m} = 18500 \text{ J}$$

(Note: We do not need to calculate the circumference because we need to take displacement and not distance because displacement is change of position not distance covered by the object)

$$\begin{aligned}
 50. F_{\text{gravitation}} &= \frac{G \times M_e \times M_s}{(\text{Dist. of the earth from the sun})^2} \\
 &= \frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 2 \times 10^{30}}{1.5 \times 10^{11} \times 1.5 \times 10^{11}} \\
 &= \frac{6.67 \times 6 \times 2 \times 10^{30-11+24-11-11}}{2.25} \\
 &= 35.57 \times 10^{21} \text{ N}
 \end{aligned}$$

51. (a) Social value, conceptual value, presence of mind.

(b) Potential energy.

52. a. chemical energy of food converts to heat and then to muscular energy which converts into kinetic energy on paddling the cycle.

b. Anil wisely opts for the bicycle as he was concerned towards environment, to avoid noise and air pollution. Also cycling is beneficial for health.

c. By informing them about the harms of pollution and putting forward the advantages of cycling.

53. If we place an iron nail on the surface of water, it sinks. This is because the density of iron is greater than that of water, so the weight of the nail is more than the upthrust of water on it. On the other hand a ship made of iron does not sink. This is because the ship is hollow and the empty space contains air which makes the average density of the ship less than that of water. Therefore, even with a small part of it submerged into water, the weight of the water displaced becomes equal to the total weight of the ship and hence the ship floats.

54. Weight of an object = mg

where 'm' is mass of the object at the equator than at the poles as the magnitude of 'g' is less at the equator than at the poles.

So, his friend will not agree with weight of the gold at the poles when measured at equator.

55. Power of Avinash $P_A = F_A \cdot v_A = 10 \times 8 = 80 \text{ W}$

Power of Kapil $P_K = F_K \cdot v_K = 25 \times 3 = 75 \text{ W}$

So, Avinash is more powerful than Kapil.

56. i. Work is said to be done only when the point of application of a force moves in the direction of the applied force.

Mathematically, work = Force $\times$ Displacement.

It is a scalar quantity and its SI unit is the joule (J).

Work done by the force of friction is negative because this force is acting opposite to the direction of displacement.

ii. Yes, a body can have energy without having momentum when a stone lying on a roof a building then its momentum is zero but energy is stored in the form of potential energy in the stone.

57. When the pendulum bob is pulled (say towards left), the energy supplied is stored in it is the form of P.E. on account of its higher position. When the pendulum is released so that it starts moving towards right, then its P.E. changes into K.E., such that in mean position, it has maximum K.E. and zero P.E. As the pendulum moves towards extreme right, its K.E. changes into P.E. and zero K.E. When it moves from this extreme position to mean position, its P.E. again changes to K.E. This illustrates the law of conservation of energy. Eventually, the bob comes to rest, because during each oscillation a part of the energy possessed by it transferred to air and in overcoming friction at the point of suspension. Thus, the energy of the pendulum is dissipated in air. The law of conservation of energy is not violated because the energy merely changes its form and is not destroyed.

58. Kinetic energy of the body is defined as the energy possessed by the virtue of motion of the body. Let a body of mass 'm' is at rest.

A force $F \text{ N}$ acts on it and produces an acceleration of a m/s^2 and it acquires a velocity of $v \text{ m/s}$ and moves a distance of $s \text{ meter}$.

Now, from third equation of motion

$$v^2 = u^2 + 2as$$

$$v^2 = 0 + 2as$$

$$a = \frac{v^2}{2s} \quad (u = 0 \text{ because, body starts from rest})$$

From Newton's second law,

$$F = ma$$

put value of 'a' from above equation

$$F = \frac{m \times v^2}{2s} = \frac{mv^2}{2s}$$

Work Done on the moving Body = Kinetic energy

$$= \text{Force} \times \text{Distance}$$

$$= \frac{mv^2}{2s} \times s = \frac{1}{2}mv^2$$

$$\text{Mass} = 5 \text{ Kg}$$

$$\text{Velocity} = 2.5 \text{ m/s}$$

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 5 \times (2.5)^2$$

$$= \frac{3125}{2 \times 100}$$

$$= \frac{1562.5}{100}$$

$$\text{K. E} = 15.625 \text{ J}$$

59. For the vertically upward motion of the ball

$$\text{Initial velocity (u)} = 20 \text{ ms}^{-1}$$

$$\text{Final velocity (v)} = 0$$

$$\text{Time (t)} = ?$$

$$\text{Distance covered upward (S)} = ?$$

$$\text{Acceleration due to gravity (g)} = -10 \text{ ms}^{-2}$$

For finding time taken by the ball to move upwards

$$\text{Using } v = u + gt$$

$$\Rightarrow 0 = 20 - 10 \times t$$

$$\Rightarrow 10t = 20$$

$$\Rightarrow t = 2\text{s}$$

As the ball takes 2 s to attain the maximum height, therefore, it will take another 2 s to just pass the thrower (since time of ascent is equal to the time of descent) thus, the time in which the ball passes the thrower.

$$= (2 + 2) \text{ s} = 4 \text{ s}$$

The total distance travelled by the ball in moving upwards and returning to the thrower can be obtained as follows

$$v^2 - u^2 = 2as$$

$$\Rightarrow 0^2 - (20)^2 = 2 \times (-10) \times S$$

$$\Rightarrow S = \frac{-400}{-20} = 20\text{m}$$

For the downward motion of the ball from the top we have

$$\text{Initial velocity } u = 0 \text{ ms}^{-1}$$

$$\text{Distance of free fall (S)} = 60 + 20 = 80 \text{ m}$$

$$\text{Time take } t = ?$$

$$\text{Using } S = ut + \frac{1}{2}at^2 \text{ we have}$$

$$80 = 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$\Rightarrow t^2 = 16$$

$$\Rightarrow t = 4\text{s}$$

Therefore total time taken by the ball to reach the ground level = (2 + 4) = 6 s.

60. At $v_1 = 18 \text{ kmh}^{-1} = \frac{18 \times 1000}{3600} = 5 \text{ ms}^{-1}$

$$\text{KE}_1 = \frac{1}{2}mv_1^2 = \frac{1}{2} \times 40 \times (5)^2 = 500 \text{ J}$$

$$\text{At } v_2 = 36 \text{ kmh}^{-1} = 10 \text{ ms}^{-1}$$

$$\text{KE}_2 = \frac{1}{2}mv_2^2 = \frac{1}{2} \times 40 \times (10)^2 = 2000 \text{ J}$$

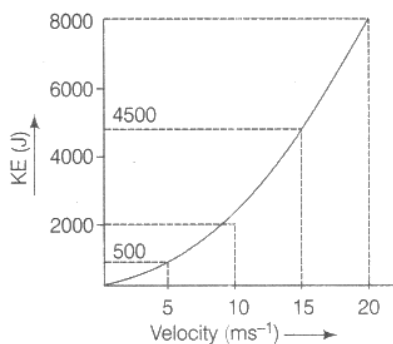
$$\text{At } v_3 = 54 \text{ kmh}^{-1} = 15 \text{ ms}^{-1}$$

$$\text{KE}_3 = \frac{1}{2}mv_3^2 = \frac{1}{2} \times 40 \times (15)^2 = 4500 \text{ J}$$

$$\text{At } v_4 = 72 \text{ kmh}^{-1} = 20 \text{ ms}^{-1}$$

$$\text{KE}_4 = \frac{1}{2}mv_4^2 = \frac{1}{2} \times 40 \times (20)^2 = 8000 \text{ J}$$

The graph between the KE and the velocities is shown below:



The graph is parabolic curve because $KE \propto v^2$.

61. i. Work is said to be done only when the point of application of a force moves in the direction of the applied force.
Mathematically, work = Force $\times$ Displacement.

Two factors on which work depends :

- The magnitude of applied force
- Displacement in the direction of the force.

- ii. Consider the masses of car and truck are $2m$ and $3m$.

Then Kinetic energy of the car = $\frac{1}{2}mv^2 = \frac{1}{2} \times 2m \times (30)^2 = 900 \text{ m}$.

Kinetic energy of the truck = $\frac{1}{2}mv^2 = \frac{1}{2} \times 3m \times (30)^2 = 1350 \text{ m}$.

The ratio of their kinetic energies = $900 \text{ m} : 1350 \text{ m} = 2 : 3$

62. i. If K is kinetic energy.

If initial kinetic energy is K_1

So, kinetic energy $K_1 = \frac{1}{2}mv^2 \dots(1)$

As per given condition if mass increased four times ($4m$) and velocity is decreased two times $\frac{v}{2}$.

So, new kinetic energy

$$K_2 = \frac{1}{2} \times 4m \times \left(\frac{v}{2}\right)^2$$

$$= 2m \times \frac{v^2}{4}$$

$$= \frac{1}{2}mv^2 \dots(2)$$

Thus from (i) and (ii)

$$K_1 : K_2 = 1 : 1$$

Hence the ratio of the kinetic energies is $1 : 1$.

- ii. When a sparkle is lighted, then chemical energy is converted into heat and light energy.

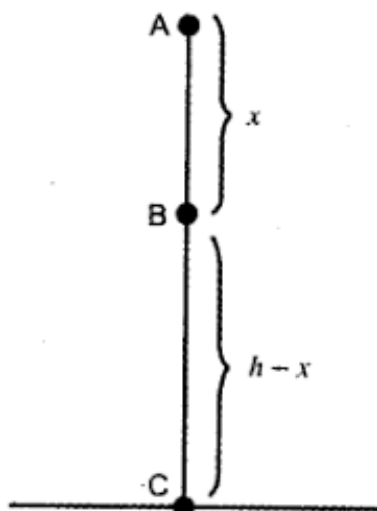
63. a.

- A cricket/hockey ball having small mass which has been hit hard and is travelling fast (possess high kinetic energy).
- A shot put having large mass thrown by an athlete possess low kinetic energy.

- b. Let the body of mass m at height, h above the ground at point A.

At point 'A'

Its K.E. = 0, P.E. = mgh



The total energy of the body at height $h = \text{P.E.} + \text{K.E.} = mgh + 0 = mgh$

At Point 'B' : After the body has fallen freely through a distance x (say) at point 'B',

Its Kinetic Energy (KE) at point 'B' = mgh and Potential Energy (PE) = $mg(h - x)$

Total energy possessed by a ball at point 'B' = $\text{KE} + \text{PE} = mgh + mg(h - x) = mgh$

At Point C: When it reaches the ground at point 'C' ;

$$v^2 - u^2 = 2aS$$

$$v^2 - 0^2 = 2gh \text{ Or, } v^2 = 2gh$$

$$\text{Its KE} = \frac{1}{2} m v^2 = \frac{1}{2} m \cdot 2gh = mgh \quad (v^2 = 2gh) \text{ and PE} = 0$$

Total Energy at point 'C' = $\text{KE} + \text{PE} = mgh$

Thus, the total mechanical energy of a freely falling body is always conserved, i.e. $(\text{KE} + \text{PE}) = mgh$.

64. Consider an undisturbed simple pendulum, in its mean position P. When the pendulum is pulled to position R, it gains height.

Thus, at position R, it has :

- i. Maximum potential energy.
- ii. Zero kinetic energy, as the pendulum is held by hand in position R.

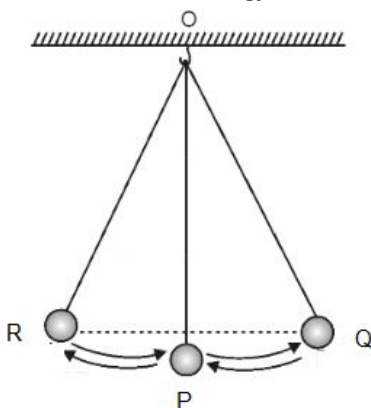
When the pendulum is released from position R, it moves towards position P. In doing so, its velocity increases.

Due to the increase in velocity, its kinetic energy increases, at the expense of potential energy. At position A, it has :

- i. Maximum kinetic energy.
- ii. Zero potential energy, as it is at its lowest position.

When the pendulum swings from P to Q, it again gain height, and hence, its potential energy increases. However, due to gain in height, its velocity decreases, and hence, the kinetic energy decreases. At position Q, it has :

- i. Maximum potential energy.
- ii. Zero kinetic energy, as pendulum comes to rest at Q for a moment, before swinging back to position P. From the above example, it is clear, that in the system of pendulum and earth, the energy is conserved. It is the potential energy, which changes into the kinetic energy and vice-versa.



65. Power of an agent which does work at the rate of 1 J per second is called 1 watt.

$$1 \text{ W} = 1 \text{ J/s}$$

$$\text{So, } 1 \text{ kW} = 1000 \text{ J/s}$$

$$\text{Mass of car} = 150 \text{ kg, power/kg} = 500 \text{ W}$$

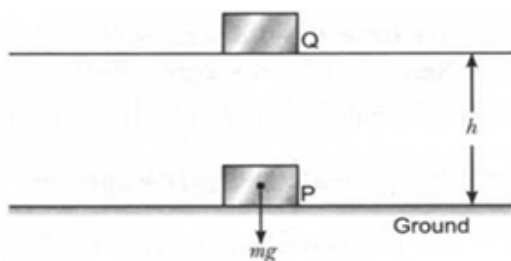
$$\text{Hence, total power} = 150 \times 500 = 7.5 \times 10^4 \text{ W}$$

We know,

$$P = \frac{W}{t} = \frac{F \times s}{t} \text{ Or } 7.5 \times 10^4 = F \times \frac{20m}{1s}$$

$$\text{Or } F = \frac{7.5 \times 10^4}{20} = 3.75 \times 10^3 \text{ N} = 3750 \text{ N}$$

66. When an object is raised through a certain height above the ground, its energy increases. This is because the work is done on it, against gravity. The energy present in such an object is called gravitational potential energy. Thus, the gravitational potential energy of an object at a point above the ground is defined as the work done in raising it from the ground to that point against gravity. Consider a body of mass m lying at point P on the Earth's surface, where its potential energy is taken as zero.



As the weight, mg acts vertically downwards, so to lift the body to another position Q at a height h , we have to apply a minimum force which is equal to mg in the upward direction. Thus, work is done on the body against the force of gravity.

We know that,

Work is done, $W = Fs$ (i)

As $F = mg$ and $s = h$

Putting these values in equation (i), we get

$$W = mg \times h = mgh$$

This work done on the body is equal to the gain in energy of the body. This is the potential energy of the body.

Therefore, Potential energy $PE = mgh$

67. According to law of conservation of energy, energy is neither created nor be destroyed. It can be only changed from one form to another.

Consider a body of mass ' m ' at rest at A at an elevation ' h ' from the ground.

Total energy at A

Potential energy of the body $= mgh$

As the body is at rest, $K.E = 0$

Total energy of the body at A $= PE + KE$

$$= mgh + 0$$

$$= mgh$$

Total energy on the ground :-

Let the body fall to ground As it strikes the ground, its elevation is 0.

because, P. E. w. rt. Ground $= 0$

Let us find its velocity as it strikes the ground at B.

As it falls freely, $v_0 = 0$

$$\text{Acceleration} = -g \text{ m/s}^2$$

Distance Travelled $= -h$

$$v^2 = u^2 + 2gh$$

$$v^2 = 2(-g)(-h)$$

$$v^2 = 2gh$$

$$K.E. = \frac{1}{2}mv^2 = \frac{1}{2} \times m \times 2gh = mgh$$

Total energy of the body on ground $= KE + PE$

$$= mgh + 0$$

$$= mgh$$

Total energy at point C.

Let the ball fall through x and be at C during its fall.

Elevation of the body at c $= h - x$

Potential energy at C $= mg(h - x)$

Let the velocity at C be v

$$v^2 = u^2 + 2as$$

$$v^2 = 2(-g)(-x)$$

$$= 2gx$$

$$K.E. = \frac{1}{2}mv^2 = \frac{1}{2} \times m \times 2gx = mgx$$

Total energy at C $= mg(h - x) + mgx$

$$= mgh$$

i. e total energy at all point of the fall is always same.

68. i. When a wet piece of cloth is shaken, small droplets of water fall down. This is because in the beginning, both water and the piece of cloth were at rest. When the piece of cloth is shaken, it is set into motion. The water tends to remain in the state of

rest due to the inertia of rest. As a result, waterfalls down in the form of droplets.

ii. $p = 100 \text{ W}$, $t = 4 \text{ h}$

Thus energy consumed = p

$$t = 100 \text{ W} \times 4 \text{ h} = 400 \text{ W h} = 0.4 \text{ kWh}$$

$$\text{Now } 1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

$$\Rightarrow 0.4 \text{ kWh} = 0.4 \times 3.6 \times 10^6$$

$$= 1.44 \times 10^6 \text{ J}$$

69. Work will be done if a force acts on an object and displacement occurs in the direction of force. According to this explanation work is done in following activities:

- Suma is doing work. She is applying force to move horizontally.
- Donkey is not doing any work. Here, the displacement and the force are at 90° .
- Work is done by the windmill. The water is lifted against force of gravity.
- No work is done by a green plant during photosynthesis.
- The engine applies a pulling force on the train, and the train moves in the direction of this force. Therefore, engine is doing work.
- During drying of food grains in the sun no work is done.
- Work is done by the air. The sailboat moves in the direction of the force exerted by wind.

70. i. The energy possessed by a body by virtue of its motion is called its kinetic energy.

No, the kinetic energy of an object cannot be negative because both m and v^2 are always positive and $KE = \frac{1}{2}mv^2$

ii. Given, mass of car $m = 1200 \text{ kg}$, Displacement covered, $s = 40 \text{ m}$, time taken, $t = 5 \text{ s}$, initial velocity, $u = 0$, $W = ?$

We know that, $W = Fs = ma \times s$ [$\because F = ma$]

$$\text{Now, } s = ut + \frac{1}{2}at^2$$

$$40 = 0 \times t + \frac{1}{2} \times a \times (5)^2$$

$$40 = \frac{1}{2} \times a \times 25$$

$$a = \frac{40 \times 2}{25} = 3.2 \text{ ms}^{-2}$$

$$\therefore W = Fs = ma \times s$$

$$= 1200 \times 3.2 \times 40 = 153600 \text{ J}$$

The work needed to be done by the car engine = 153600 J

71. i) Let V_1 = Total volume immersed (upto mid-point)

$$\begin{aligned} &= \text{Volume of the bulb} + \text{Volume of the immersed part of stem} \\ &= 12 + 0.32 \times 10 (\text{Volume} = \text{Area of cross section} \times \text{immersed length}) \\ &= 15.2 \text{ cm}^3 \end{aligned}$$

For a floating body,

Weight of solid (mg) = Upthrust (Vdg)

$$22.8 \times g = 15.2 \times d_1 \times g$$

$$\text{or } d_1 = \frac{22.8}{15.2} = 1.5 \text{ g cm}^{-3}$$

ii) V_2 = Total volume immersed = Volume of the bulb + total volume of stem

$$= 12 + 0.32 \times 20 = 18.4 \text{ cm}^3$$

Since weight = upthrust, therefore

$$22.8 = 18.4 \times d_2$$

$$\text{or } d_2 = \frac{22.8}{18.4} = 1.24 \text{ g cm}^{-3}$$

72. Electric energy consumed per day by 4 bulbs = $4 \times 40 \times 5 = 800 \text{ Wh}$

Electric energy consumed per day by 4 lights = $4 \times 60 \times 5 = 1200 \text{ Wh}$

Electric energy consumed per day by TV = $100 \times 6 = 600 \text{ Wh}$

Electric energy consumed per day by washing machine = $400 \times 3 = 1200 \text{ Wh}$

$\therefore$ Total electric energy consumed by all electric appliances = $(800 + 1200 + 600 + 1200) = 3800 \text{ Wh} = 3.8 \text{ kWh}$

Total electric energy consumed by all electric appliances = 3.8 units

Total electric energy consumed in the month of April (30 days) = $3.8 \times 30 = 114 \text{ units}$

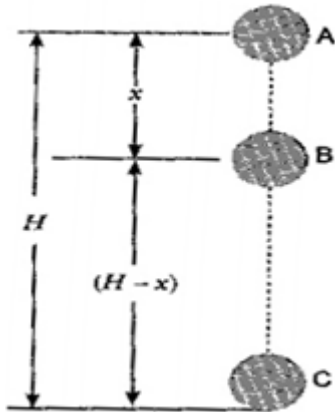
Cost of one unit = Rs.1.80

Cost of 114 units = $114 \times 1.80 = \text{Rs.}205.20$

$\therefore$ Electricity bill amount = Rs.205.20

73. The mechanical energy (kinetic energy + potential energy) of a freely falling object remains constant. It may be shown by calculation as follows:

Suppose a body of mass m falls from point A, which is at height 'H' from the surface of the earth. Initially, at point A, kinetic energy is zero and the body has only potential energy.



The total energy of the body at point A = Kinetic energy + Potential energy
 $= 0 + mgH = mgH$ (i)

Suppose during fall, the body is at position B. The body has fallen at a distance x from its initial position. If the velocity of the body at B is v , then from the formula

$$v^2 = u^2 + 2as, \text{ we have}$$

$$v^2 = 0 + 2gx = 2gx$$

Therefore, the Kinetic energy of the body at point B = $\frac{1}{2}mv^2$

$$\frac{1}{2}m \times 2gx = mgx$$

Potential energy of body at point B = $mg(H-x)$

Therefore, Total energy of a body at point B = Kinetic energy + Potential energy
 $= mgx + mg(H-x) = mgH$ (ii)

Now suppose the body is at point C, just above the surface of the earth (i.e., just about to strike the earth). Its potential energy is zero.

The height by which the body falls = H. If v is the velocity of the body at C, then from the formula

$$v^2 = u^2 + 2as$$

We have $u = 0$, $a = g$, $s = H$

$$\text{So, } v^2 = 0 + 2gH = 2gH$$

Therefore, the kinetic energy of the body at a position C = $\frac{1}{2}mv^2$

$$= \frac{1}{2}m \times 2gH = mgH$$

Therefore, Total energy of a body at C

= Kinetic energy + Potential energy

$$= mgH + 0 = mgH$$
 (iii)

Thus, we see that the sum of kinetic energy and potential energy of the freely falling body at each point remains constant.

Thus, under the force of gravity, the total mechanical energy of the body remains constant.